Brandon Miner

San Francisco, CA

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Summary Taglines (pick one per derivative)

AI Engineer — Multi-modal pipelines, LLM-assisted labeling, and production inference for unstructured video and text *[ai-engineer]*

Data Scientist — Statistical modeling, NLP pipelines, and end-to-end ML products in civic and public-interest contexts *[data-scientist]*

Forward Deployed Engineer — End-to-end ML systems on customer infrastructure — from stakeholder needs through production deployment *[forward-deployed-engineer]*

Public Technologist — Data systems for civil liberties, government accountability, and privacy advocacy *[public-technologist]*

Forward Deployed ML Engineer — Deploying production AI systems at the intersection of machine learning, infrastructure, and mission-critical operations *[forward-deployed-engineer, ai-engineer, commented]*

Machine Learning Engineer with 2 years of experience in machine learning, cloud computing, and data systems *[ai-engineer, data-scientist, commented]*

Public Interest Data Analyst with experience utilizing public data *[public-technologist, data-scientist, commented]*

Education

University of San Francisco — **San Francisco, CA**

M.S. in Data Science and Artificial Intelligence — *Expected Jun 2026*

- *Distributed Computing [data-scientist, forward-deployed-engineer, ai-engineer]*
- *Machine Learning [ai-engineer, data-scientist, forward-deployed-engineer, public-technologist]*
- *Communication [public-technologist, forward-deployed-engineer, data-scientist]*
- *Deep Learning [ai-engineer, data-scientist, public-technologist]*
- *Experiments* / Experiments in Data Science *[data-scientist, public-technologist]*
- Mlops *[ai-engineer, forward-deployed-engineer, public-technologist]*
- Computer Science for Data Science *[public-technologist, ai-engineer]*

San Diego State University — **San Diego, CA**

B.S. in Statistics with Emphasis in Data Science, Minor in Mathematics — *Dec 2025*

Experience

ACLU — role title variants — **San Francisco, CA**

AI/ML Engineer; Data Scientist; Machine Learning Engineer — *Oct 2025 – Present*

- Architecting an end-to-end ML video auditing platform from scratch (**Python, FastAPI, MySQL, Docker**) to automate analysis of unstructured police bodycam footage, projected to eliminate **300+ hours** of manual review per monthly cycle. *[data-scientist, forward-deployed-engineer, public-technologist]*
- Architecting the product development of an end-to-end ML video auditing platform from scratch (**Python, FastAPI, MySQL, Docker**) to automate data processing and analysis of unstructured police bodycam footage, projected to eliminate **300+ hours** of manual review per monthly cycle. *[data-scientist, public-technologist]*
- Architecting the software design of an end-to-end **multi-modal AI** video auditing platform from scratch (**Python, FastAPI, MySQL, Docker**) to automate analysis of unstructured police bodycam footage, projected to eliminate **300+ hours** of manual review per monthly cycle, all while tracking product development. *[ai-engineer, forward-deployed-engineer]*
- Architecting an end-to-end data pipeline ML video auditing platform from scratch (**Python, FastAPI, MySQL, Docker**) to automate analysis of unstructured police bodycam footage, projected to eliminate **300+ hours** of manual review per monthly cycle. *[data-scientist, ai-engineer]*
- Designed and built an end-to-end **ML inference platform** from scratch (**Python, FastAPI, MySQL, Docker**) to automate analysis of unstructured police bodycam footage, projected to eliminate **300+ hours** of manual review per monthly cycle. *[ai-engineer, data-scientist]*
- Owned end-to-end delivery of a production **ML inference platform** (**Python, FastAPI, MySQL, Docker**) deployed directly on client hardware — from requirements gathering through deployment — eliminating **300+ hours** of manual review per monthly cycle. *[forward-deployed-engineer]*

- Followed agile development methodologies to create an end-to-end ML video auditing platform (**Python, FastAPI, MySQL, Docker**) to automate settlement requirements review of police bodycam footage, projected to eliminate **300+ hours** of manual review per monthly cycle and expand oversight capacity to amplify the criminal justice system. *[public-technologist]*
- Developing a GPU-accelerated ETL pipeline using **FFmpeg** and **Whisper**-based speech-to-text to ingest, transcribe, and process batches of **600+ videos per month**, with **LLM-driven auto-labeling** for fine-tuning multi-stage text classification to extract actionable audit signals. *[ai-engineer, data-scientist, forward-deployed-engineer]*
- Developing a GPU-accelerated **ETL pipeline** using **FFmpeg** and **Whisper**-based speech-to-text to ingest, transcribe, and process batches of **600+ videos per month**; integrated **LLM-driven auto-labeling** to generate fine-tuning data for multi-stage text classification models extracting actionable audit signals. *[data-scientist, ai-engineer]*
- Engineered a **GPU-accelerated inference pipeline** using **FFmpeg** and **OpenAI Whisper** for speech-to-text transcription, processing **600+ videos per month**; integrated **LLM-driven auto-labeling** to generate training data for fine-tuning multi-stage text classification models. *[ai-engineer]*
- Engineered a **GPU-accelerated ETL and inference pipeline** using **FFmpeg** and **Whisper ASR** for batch transcription of **600+ videos per month**; developed an **LLM-driven auto-labeling** system to generate training data for fine-tuning multi-stage text classification models. *[ai-engineer, data-scientist]*
- Embedded with non-technical stakeholders to translate operational needs into a **GPU-accelerated ETL and inference pipeline** using **FFmpeg** and **Whisper ASR**, processing **600+ videos per month** with zero cloud dependency. *[forward-deployed-engineer]*
- Developing a GPU-accelerated ETL pipeline using **FFmpeg** and **Whisper**-based speech-to-text to ingest, transcribe, and process **600+ police videos per month**, with **LLM-driven auto-labeling** to flag potential rights violations and misconduct patterns for legal review. *[public-technologist]*
- Designed and trained **multi-stage NLP classifiers** on auto-labeled transcripts to detect use-of-force indicators, rights violations, and officer conduct patterns at scale. *[data-scientist, public-technologist]*
- Architected a **multi-stage NLP classification system** with LLM-assisted labeling, iterative fine-tuning, and structured output parsing to extract actionable audit signals from raw transcripts at scale. *[ai-engineer, data-scientist]*
- Architected and shipped a **multi-stage NLP classification system** combining LLM-assisted auto-labeling, iterative fine-tuning, and structured output parsing — delivering audit-ready signals directly in the hands of end-users. *[forward-deployed-engineer, ai-engineer]*
- Building a **private-server FastAPI backend** and **React.js** web frontend delivering the full auditing system to non-technical end-users. *[forward-deployed-engineer, ai-engineer]*
- Built a **private-server FastAPI backend** and **React.js** web frontend to deliver the full auditing system to non-technical stakeholders, enabling self-service model inference without engineering overhead. *[data-scientist, forward-deployed-engineer, public-technologist]*
- Designed and deployed a **private-server FastAPI backend** paired with a **React.js** frontend, delivering the full AI auditing system as a production-ready web application for non-technical end-users. *[ai-engineer, forward-deployed-engineer]*
- Deployed the full ML system on a **private-server FastAPI backend** with a **React.js** web frontend, enabling reliable model serving and inference for non-technical end-users without cloud dependency. *[ai-engineer, forward-deployed-engineer]*
- Deployed the full ML stack on a **private-server FastAPI backend** with a **React.js** web frontend, enabling non-technical civil rights investigators to run inference without any engineering support. *[forward-deployed-engineer, public-technologist]*
- Delivering the full auditing system to non-technical legal and advocacy staff via a **private-server FastAPI backend** and **React.js** web frontend, ensuring sensitive footage remains secure and off commercial infrastructure. *[public-technologist]*
- Leading a **geospatial privacy analysis** initiative to expose deanonymization risks in commercially sold location data, directly informing advocacy efforts for legislative bans on data broker practices. *[public-technologist]*
- Pushing a **geospatial** privacy analysis initiative to expose deanonymization risks in commercially sold location data, directly supporting advocacy efforts for legislative bans. *[public-technologist, data-scientist, commented]*
- Engineered a spatial data pipeline using **Plotly**, OpenStreetMap, and the **Google Reverse Geocoding API** to translate raw GPS coordinates into identifiable behavioral patterns, applying **DBSCAN clustering** to detect sensitive personal locations (home, work, medical facilities) from purportedly anonymized data, producing concrete evidentiary exhibits for policy arguments. *[public-technologist]*
- Engineered a spatial data pipeline using **Plotly**, OpenStreetMap, and the **Google Reverse Geocoding API** to translate raw GPS coordinates into identifiable behavioral patterns, applying **DBSCAN clustering** to automatically detect sensitive personal hotspots (home, work, medical facilities) from anonymized location traces. *[public-technologist, data-scientist, commented]*

San Diego County Taxpayers Association — **San Diego, CA**
Data Scientist, Lead Intern — *Mar 2024 – Aug 2025*

- Led data science initiatives, managing a team of **4+ associates** to deliver actionable insights on civil workforce and homelessness programs. *[data-scientist, public-technologist]*
- Led data science initiatives, managing a team of **4+ associates** to run statistical data analysis to deliver actionable ML-driven insights on civil workforce and homelessness programs, utilizing leadership and communication skills. *[ai-engineer, data-scientist]*
- Led data science initiatives, managing a team of **4+ associates** to deliver statistical insights on civil workforce economics and regional homelessness programs using statistical data analysis. *[data-scientist]*
- Learned team leadership through statistical analysis initiatives I led across collaborative civic policy research, managing a team of **4+ associates** to deliver quantitative analysis on public workforce, government spending, and homelessness programs for public reporting. *[public-technologist]*
- Led a team of **4 analysts** to build data pipelines delivering statistical analysis of civil workforce compensation and homelessness intervention programs. *[data-scientist, ai-engineer]*
- Led a cross-functional team of **4 analysts** to build data pipelines and deliver analytical products directly to government stakeholders, translating messy public datasets into actionable policy insights. *[forward-deployed-engineer, public-technologist]*
- Built a public workforce analytics pipeline using iterative **web scraping** and **NLP-based standardization** to analyze wage dynamics, identifying significant taxpayer cost of **19.7%** police wage increase per **1%** staffing drop). *[data-scientist, public-technologist]*
- Built a workforce analytics pipeline combining iterative **web scraping** and **NLP-based entity standardization** to analyze wage dynamics across public records, surfacing a statistically significant **19.7%** police wage increase per **1%** staffing drop. *[ai-engineer, data-scientist]*
- Built a public workforce analytics pipeline combining **iterative web scraping** and **NLP-based entity standardization** to clean and link heterogeneous salary records; applied regression modeling to quantify a **19.7% police wage increase per 1% staffing drop**. *[data-scientist]*
- Built a public workforce analytics pipeline using iterative **web scraping** and **NLP-based standardization** to analyze wage dynamics in civil service, surfacing a **19.7% police wage increase** per **1% staffing drop**. *[public-technologist]*
- Developed an **data standardization pipeline** combining iterative web scraping and text normalization to unify messy public wage records; applied regression modeling to quantify a **19.7% police wage increase** per **1%** staffing reduction. *[data-scientist, ai-engineer]*
- Designed and deployed an end-to-end **data standardization pipeline** combining iterative web scraping and text normalization; applied regression modeling to quantify a **19.7% police wage increase** per **1%** staffing reduction — findings presented to county officials. *[forward-deployed-engineer, public-technologist]*
- Analyzed regional homelessness expenditure using **mixed-effects models** and **multicollinearity reduction**, identifying high-impact interventions like rapid rehousing for future resource allocation. *[data-scientist, public-technologist]*
- Applied **mixed-effects regression models** and **multicollinearity reduction** to analyze regional homelessness expenditure data, identifying high-impact interventions (e.g., rapid rehousing) to guide future resource allocation decisions. *[ai-engineer, data-scientist]*
- Modeled regional homelessness expenditure using **mixed-effects regression** and **multicollinearity reduction (VIF/PCA)**, isolating high-impact interventions such as rapid rehousing to guide future resource allocation recommendations. *[data-scientist]*
- Applied **mixed-effects regression models** with multicollinearity reduction (VIF analysis) to regional homelessness expenditure data, surfacing high-impact interventions such as rapid rehousing to guide resource allocation decisions. *[data-scientist, ai-engineer]*
- Applied **mixed-effects regression** with VIF-based multicollinearity reduction to regional homelessness expenditure data; surfaced high-impact interventions to guide resource allocation decisions for program leadership. *[forward-deployed-engineer, data-scientist]*
- Analyzed regional homelessness expenditure using **mixed-effects models** and **multicollinearity reduction** to evaluate program efficacy, identifying high-impact interventions like rapid rehousing to guide future public resource allocation. *[public-technologist]*

Technical Projects

GenAI Franchise Personality Quiz

- Built an end-to-end quiz generator: users describe any fictional class system in natural language; Gemini parses the franchise, scrapes Fandom wikis for canon characters, and roleplays each through 15 Likert questions to produce reference vectors asynchronously via FastAPI background jobs. *[ai-engineer, forward-deployed-engineer]*
- Implemented a vector-based classifier that scores user answers with cosine similarity against LLM-generated character embeddings, ranks classes by per-type mean similarity, and projects results into 3D PCA for interactive visualization in a Next.js results UI. *[ai-engineer, data-scientist]*

- Deployed a split architecture on Vercel (frontend) and Google Cloud Run (API), with durable quiz artifacts in GCS, shareable quiz URLs, a community quiz library, and GitHub Actions CI (pre-commit, pytest, automated API deploy). *[ai-engineer, forward-deployed-engineer]*
- Designed a multi-tier quiz resolver (in-memory → local disk → GCS) so generated quizzes survive API restarts and can be shared and retaken without re-running LLM generation. *[ai-engineer, forward-deployed-engineer]*
- Hardened production behavior with per-IP daily rate limiting, structured JSON logging on Cloud Run, user-facing handling of Gemini 429/503 overload, and a pytest + Playwright test suite (90 unit/functional tests plus opt-in full-stack E2E). *[ai-engineer, forward-deployed-engineer]*

California Grape ETL Pipeline & Dashboard (title variants below)

- Project title variants: *California Grape ETL Pipeline & Dashboard*; *& Cloud Dashboard*; *& Analytics Dashboard*; *& Public Dashboard*.
- **Engineered a multi-source ETL pipeline** in Python to extract, transform, and consolidate disparate agricultural datasets into a unified schema for analysis. *[data-scientist, forward-deployed-engineer]*
- **Engineered a multi-source ETL pipeline** in Python to extract, transform, and consolidate disparate agricultural datasets into a unified schema for analysis and data modeling. *[data-scientist]*
- **Engineered a multi-source ETL pipeline** in Python to extract, transform, and consolidate disparate agricultural datasets into a unified schema for downstream ML and analytics. *[ai-engineer, data-scientist]*
- Engineered a **multi-source ETL pipeline** in Python to extract, transform, and consolidate disparate agricultural datasets into a unified schema optimized for downstream analysis. *[data-scientist]*
- Engineered a **multi-source ETL pipeline** in Python to extract, transform, and consolidate disparate agricultural datasets into a unified schema ready for downstream ML and analytics workloads. *[ai-engineer, forward-deployed-engineer]*
- Built a **multi-source ETL pipeline** in Python to extract, transform, and consolidate disparate agricultural datasets into a unified schema, demonstrating ability to rapidly onboard domain-specific data and deliver working systems. *[forward-deployed-engineer]*
- **Engineered a multi-source ETL pipeline** in Python to extract, transform, and consolidate disparate agricultural datasets into a unified schema, making government-held data accessible for public analysis. *[public-technologist]*
- **Containerized the full-stack application** using Docker Compose and deployed a serverless instance to **Google Cloud Run**, ensuring scalable and reliable cloud hosting. *[ai-engineer, forward-deployed-engineer]*
- **Containerized the full-stack application** with **Docker Compose** and deployed to **Google Cloud Run** as a scalable serverless instance — reproducible infrastructure ready for client-site or cloud deployment. *[forward-deployed-engineer]*
- **Containerized the full-stack application** using **Docker Compose** and deployed to **Google Cloud Run** as a serverless instance, enabling scalable, reproducible model and app serving. *[ai-engineer, forward-deployed-engineer]*
- **Containerized and deployed a technical delivery** via Docker Compose and **Google Cloud Run**, ensuring reliable public access to a **live interactive Streamlit dashboard** for data visualization of regional production data. *[public-technologist]*
- **Developed an interactive Streamlit front-end** dashboard, enabling end-users to dynamically visualize and filter regional production metrics and trends. *[data-scientist, public-technologist]*
- **Developed an interactive Streamlit front-end** dashboard for real-time filtering and visualization of regional production metrics. *[data-scientist]*
- Built an interactive **Streamlit front-end** dashboard for dynamic visualization and filtering of regional production metrics; demonstrates end-to-end deployment from pipeline to user-facing interface. *[ai-engineer, forward-deployed-engineer]*
- Delivered an interactive **Streamlit dashboard** for dynamic visualization of regional production metrics, providing a user-facing interface accessible to non-technical stakeholders. *[forward-deployed-engineer, public-technologist]*
- Developed an **interactive Streamlit dashboard** enabling end-users to dynamically visualize and filter regional production metrics; view live deployment. *[data-scientist, public-technologist]*
- View Live Dashboard

Parkcast-SF: San Francisco Parking Assistant

- Owned system design and the production **FastAPI** inference API for a full-stack USF capstone parking platform, serving block-level **LightGBM** occupancy predictions from artifacts in **GCS**. *[ai-engineer, forward-deployed-engineer]*
- Applied **K-nearest neighbors**-derived baselines to support **LightGBM** inference on sparse, non-metered blocks where direct sensor history was unavailable. *[ai-engineer, forward-deployed-engineer]*
- Built automated model **CI/CD** on **GCP**: scheduled **GitHub Actions** retraining with **MLflow** experiment tracking and a **champion/challenger** promotion gate (validation MAE) before GCS upload; split deploy across **Vercel** (frontend) and **Cloud Run** (API). *[ai-engineer, forward-deployed-engineer]*
- Designed the full system architecture for an urban parking assistant app using historical meter data, location-based modeling, and production-ready MLOps. *[ai-engineer, forward-deployed-engineer, commented]*
- Built a Vercel frontend to interact with users and a Google Cloud Run API entrypoint to serve backend predictions and handle app traffic. *[ai-engineer, forward-deployed-engineer, commented]*

- Predicted parking availability across SF using historical parking meter records; where meters were unavailable, applied a **K-nearest neighbors** approximation to infer nearby parking availability. *[data-scientist, ai-engineer, commented]*
- Implemented a challenger/champion model framework with a new challenger deployed every 3 days to continuously test and improve production performance. *[ai-engineer, forward-deployed-engineer, commented]*
- Used OpenStreetMap and Mapbox to simulate Google Maps mapping experience and support navigation/visualization without Google Maps dependency. *[forward-deployed-engineer, commented]*
- Automated data ingestion into a Google Cloud Storage data lake using GitHub Actions, supporting reproducible model retraining and feature updates. *[ai-engineer, forward-deployed-engineer, commented]*
- Created a separate Cloud Run instance to host **MLflow** for upstream challenger/champion experiment tracking and model evaluation. *[ai-engineer, data-scientist, commented]*

Credit Card Fraud Detection (title variants below)

- Project title variants: *Credit Card Fraud Detection — Optimized Classification Model*; *Credit Card Fraud Prediction Model*; *Threshold-Optimized Classification*; *Imbalanced Classification.*
- Built a binary classification model using **Logistic Regression** (**Scikit-Learn**) with **PCA**-based dimensionality reduction to isolate the 16 highest-variance features for fraud detection. *[ai-engineer, data-scientist]*
- Built a binary classification model using Logistic Regression (**Scikit-Learn**) to identify fraudulent credit card transactions. *[data-scientist, commented]*
- Built a binary classification model using **Logistic Regression (Scikit-Learn)** to identify fraudulent transactions on a highly imbalanced dataset. *[data-scientist]*
- Trained a binary classification model (**Logistic Regression, Scikit-Learn**) on highly imbalanced transaction data to detect fraudulent activity with production-grade precision-recall constraints. *[ai-engineer, data-scientist, commented]*
- Applied **Principal Component Analysis (PCA)** for dimensionality reduction and multicollinearity mitigation, successfully isolating the 16 highest-variance features for model training. *[data-scientist, commented]*
- Applied **PCA** for dimensionality reduction and multicollinearity mitigation, isolating the **16 highest-variance features** to improve model generalization and reduce training complexity. *[data-scientist, ai-engineer, commented]*
- Performed **threshold optimization** via ROC-AUC analysis to minimize False Negative Rate while strictly capping False Positive Rate at **under 2%**, directly maximizing business value. *[ai-engineer, data-scientist]*
- Analyzed the **ROC-AUC tradeoff** to dynamically adjust the classification threshold, intentionally minimizing the False Negative Rate to catch undetected fraud while strictly capping the False Positive Rate at **under 2%**. *[data-scientist, commented]*
- Tuned classification threshold via **ROC-AUC analysis** to minimize False Negative Rate (missed fraud) while capping False Positive Rate at **under 2%**. *[data-scientist]*
- Performed **ROC-AUC threshold optimization** to dynamically tune the classification boundary, minimizing False Negatives to maximize fraud recall while enforcing a strict **False Positive Rate cap under 2%** — directly modeling a real-world business constraint. *[ai-engineer, data-scientist, commented]*
- View blog post / full write-up

Technical Skills

Programming Languages & Frameworks: ML software engineering, object-oriented programming, Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, FastAPI, Streamlit, Transformers, Hugging Face), SQL (PostgreSQL), R (tidyverse), Statistical Software, Bash, Spark SQL, Apache Spark, Javascript (React.js)
Data Engineering: multi-source ETL pipelines, data standardization, web scraping, text normalization, entity standardization, BigQuery, MongoDB, geospatial data pipelines
AI / ML Algorithms & Methods: Deep Learning (Neural Networks, CNNs, Transformers, fine-tuning), LLM integration & prompting, LLM-driven auto-labeling, NLP (ASR, speech-to-text, text classification, NER, multi-stage classifiers), Regression, mixed-effects regression, Ensemble Methods (Random Forests, XGBoost, AdaBoost), Bayesian text classification, PCA, multicollinearity reduction (VIF), ROC-AUC threshold optimization, imbalanced classification, Recommender systems, Explainable AI (XAI), Geospatial Analysis (DBSCAN, OpenStreetMap, Plotly), distributed computing (Spark), multi-modal AI
MLOps, Deployment & Infrastructure: Docker, Docker Compose, Google Cloud Platform (GCP), Google Cloud Run, cloud computing, serverless deployment, GPU-accelerated inference, REST APIs (FastAPI), private-server deployment, client-hardware deployment, Git, GitHub, version control systems, agile development, product development tracking
Tools & Integrations: OpenAI Whisper, FFmpeg, LLM-based labeling, web scraping, Google Reverse Geocoding API, PostgreSQL
Soft / Cross-functional: team leadership, stakeholder embedding, requirements gathering, communication, civic policy research, legal/advocacy delivery